EBU6304 Software Engineering Group Project

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First report

1. Requirements

1.1 Fact-finding approach

We used a **triangulated fact-finding** process so that the backlog was based on real workflow problems rather than only on the handout. First, we analysed the coursework brief to identify mandatory technical constraints, required roles, and the minimum scope expected in the first assessment. In this report, we explain our fact-finding techniques, iteration plan, prioritisation method and estimation method, and confirmed that the product itself must be either a Java desktop application or a lightweight Servlet/JSP web system with file-based storage rather than a database.

Second, we studied the **existing recruitment workflow** and converted it into an “as-is” process summary. From the supporting materials, the current practice already contains digital elements, but it is fragmented across the university information portal, an online form and separate CV submission by email. This fragmentation created repeated context switching for applicants and repeated manual reconstruction of candidate information for organisers.

Third, we conducted **semi-structured interviews** with three stakeholder perspectives: TA applicants, module organisers/reviewers, and administrative/co-ordination staff. We chose semi-structured interviews because they let us test our assumptions while still allowing interviewees to explain pain points in their own words. This was important because the problem is not purely technical; it depends heavily on how users experience clarity, fairness and review efficiency. Interview findings were then coded into recurring themes such as **“integrated workflow”, “status visibility”, “clear vacancy requirements”, and “structured applicant comparison”**.

Fourth, we ran a **pilot survey** with 24 respondents to validate whether the interview themes were broad enough to guide prioritisation. The survey gave us a quick quantitative check. The strongest signals were the need to reduce switching between portal/form/email, better application status tracking, clearer vacancy details, and integrated in-system submission. By contrast, AI-style features such as skill matching and missing-skill hints were seen as useful but clearly secondary.

Finally, we used **prototype-based validation**. Our wireframes cover the main role-specific flows for TA, MO and Admin, including login, registration, vacancy browsing, vacancy details, application submission, application tracking, job posting, applicant review and workload overview. Mapping pages to user stories helped us verify that the prototype and backlog were aligned before implementation.

1.2 Main requirements identified from the evidence

The evidence consistently showed that the current recruitment process is not failing because there are no online tools; it fails because the workflow is split across multiple channels. Therefore, the first requirement of the system is **integration**: applicants should be able to browse vacancies, maintain a profile, upload a CV, submit an application and later check status within one system. The most valuable first-release outcome is workflow completeness rather than feature novelty.

A second requirement is **transparency**. Applicants want to know what each vacancy really involves before they apply, and they want feedback after submission. This led us to emphasise a detailed vacancy page, clear required skills/workload information, and a visible application status mechanism. Survey results showed that detailed vacancy information and status tracking were among the highest-rated functions.

A third requirement is **reviewer efficiency**. Module organisers need structured data that can be compared quickly while still retaining access to the original CV as evidence. This motivated organiser stories around posting vacancies in a structured format, viewing applicants in one place, and updating outcomes centrally. The evidence suggests that comparison support matters more than automatic ranking.

A fourth requirement is **responsible use of AI**. Both interview and survey feedback showed that matching-related functions would only be acceptable if they are explainable and clearly advisory. For that reason, AI-assisted recommendation and missing-skill highlighting were deliberately moved out of the MVP and placed in a later iteration after the end-to-end recruitment workflow is stable.

1.3 Translating findings into user stories and the product backlog

After gathering evidence, we translated the findings into **Agile user stories** and then organised them into the **product backlog**. Following the approach introduced in the course materials, each story was written in the form "**As a user, I want, so that**," so that every backlog item retained a clear user role, a concrete goal, and an identifiable benefit. This helped us keep the focus on user value rather than on technical implementation details alone.

We **first identified three main user groups** from the current workflow: **TA applicants, module organisers, and admin or coordination staff**. The evidence gathered from interviews and the pilot survey was then converted into role-specific stories. For example, the repeated complaint about switching between the portal, form, and email was translated into stories for registration, CV upload, vacancy browsing, and application submission within one system. The need for more transparency was translated into stories for viewing detailed vacancy information and checking application status. Organiser concerns about inefficiency and fragmented evidence were translated into stories for structured vacancy posting and applicant review.

We also used **acceptance criteria** to make each story more precise. This was important because a short story statement alone is not enough to define what "done" means. Acceptance criteria allowed us to describe expected system behaviour in a practical and testable way, such as which fields are required, which roles can access a page, how uploaded files should be validated, or when a vacancy should no longer appear as open. This directly follows the course guidance that acceptance criteria support understanding, negotiation, and later testing.

The backlog was then structured using **story IDs, story names, descriptions, priorities, iteration numbers, acceptance criteria, and effort estimates**. During this process, we also refined story scope to keep each item implementable and testable. For example, vacancy discovery was consolidated as one story (PB-10: browse/search/filter), while vacancy details (PB-11), application submission (PB-12), and status tracking (PB-13) were kept as separate items. In the same way, AI-assisted features were kept separate from the core profile and application flow so that they could be postponed without weakening the MVP.

This conversion step was important because it created a clear **trace from evidence to implementation planning**. Rather than treating the backlog as a simple feature list, we used it as a structured record of which requirements emerged from stakeholder feedback, which stories were essential for the first release, and which enhancements could be delivered later.

1.4 Iteration plan

We planned the backlog in four iterations using dependency-first delivery.

Iteration 1 focuses on **technical foundations and core platform capabilities**: role-based login, TA registration, CV/profile support, session management, JSON-based persistence, role-based access control, and admin setup of MO accounts. These stories reduce architectural risk early and establish a usable base for later recruitment workflows. The relevant backlog items are PB-01, PB-02, PB-03, PB-04, PB-05, PB-06, PB-07, PB-08 and PB-09.

Iteration 2 completes the applicant-side recruitment journey by **enabling vacancy discovery and application handling**. It includes vacancy browsing/search/filter, vacancy details, application submission, and status tracking. The relevant backlog items are PB-10, PB-11, PB-12 and PB-13.

Iteration 3 addresses **organiser-side workflow** through MO operations once applicant and vacancy data are already stable. It covers MO profile management, vacancy posting, and applicant review. The relevant backlog items are PB-14, PB-15 and PB-16.

Iteration 4 contains **advanced and administrative enhancements** after the core workflow is complete. It includes admin workload overview, application withdrawal/revocation handling, vacancy recovery and TA revocation management, and AI-assisted suitability analysis. The relevant backlog items are PB-17, PB-18, PB-19, PB-20 and PB-21.

Overall, the iteration plan reflects incremental delivery. Early iterations focus on making the basic workflow usable and testable. Later iterations then improve management visibility and add carefully scoped intelligent assistance.

1.5 Prioritisation method

We used **MoSCoW prioritisation** and refined it through **three practical checks: user value, delivery dependency, and implementation risk.** This approach matched the course guidance that backlog items should be prioritised rather than treated as an undifferentiated feature list.

Must-have items were defined as stories without which the system could not demonstrate its core value or would fail the coursework constraints. These include login, registration, profile and CV handling, vacancy browsing, vacancy detail viewing, application submission, application status tracking, organiser vacancy posting and applicant review, role-based access control, and JSON-based persistence. Together, these stories form the minimum end-to-end recruitment workflow.

Should-have and could-have items were treated as controlled extensions. Admin workload overview is useful for broader management, but it is not essential to prove that recruitment can be completed from vacancy posting to decision update. Similarly, AI-powered vacancy suitability analysis and applicant skill-gap analysis were classified as lower priority because the handout presents them as possible AI-powered features rather than mandatory system functions, and because our own evidence showed that users preferred explainable support tools only after the basic workflow worked reliably.

Our prioritisation also **reflects release strategy**. We intentionally avoided introducing advanced matching or analytics too early, because doing so would add complexity before the system had stable data capture, permissions, and status handling. In that sense, prioritisation in this project is not only about ranking features. It is also about deciding which order best produces a usable and defensible product at each checkpoint.

1.6 Estimation method

We estimated backlog items using a **lightweight story-point approach** based on relative effort. We used Fibonacci-style values such as 3, 5, and 8 because they are suitable for early Agile planning, where uncertainty is still present and exact hour-level prediction would give a false sense of precision. This is consistent with the course materials, which present story points as a way to express effort, complexity, and uncertainty together.

In our calibration, **3-point stories are relatively self-contained items** with limited validation or interaction logic. Examples include viewing a page, checking a status, or enforcing a basic role-based access rule. **Five-point stories involve more interaction**, validation, or data handling, such as registration, profile management, vacancy posting, or application submission. **Eight-point stories involve wider workflow coordination and higher testing effort**. The clearest example is organiser-side applicant review, where profile retrieval, CV access, decision updates, and downstream status changes all need to work together correctly.

We used these estimates to support sprint-level planning rather than to maximise the number of stories delivered per iteration. This was important in a student Agile project, because each checkpoint needs to show visible working software and a sensible trade-off between ambition and feasibility. Story points therefore supported discussion and planning discipline rather than acting as exact predictions.

1.7 Summary

Overall, our report shows a clear chain from evidence to backlog. Fact-finding highlighted fragmentation, lack of transparency and inefficient review. Those findings directly shaped our MVP scope, our sprint order and our decision to postpone AI features until they can be implemented responsibly. As a result, the first release is designed to solve the most frequent and highest-value problems first while remaining compliant with the coursework constraints.

1.8 Sprint Summary Table

Iteration	Sprint goal	Key backlog items	Rationale
1	Foundation and compliant data model	PB-01, PB-02, PB-03, PB-04, PB-05, PB-06, PB-07, PB-08, PB-09	Removes technical risk early and enables a demonstrable entry flow.
2	Complete applicant journey	PB-10, PB-11, PB-12, PB-13	Delivers the highest end-user value and the first end-to-end recruitment workflow.
3	Support organiser decision-making	PB-14, PB-15, PB-16	Depends on vacancies and applications already existing in the system.
4	Add management and AI enhancements	PB-17, PB-18, PB-19, PB-20, PB-21	Useful extensions, but not required for the MVP.

2. Supporting material

2.1 Findings Based on the Existing Recruitment Process

2.1.1 Current process summary

Based on the current recruitment practice, TA vacancies are first announced through the university information portal. Applicants then complete an online form to provide basic information, select relevant modules or skill areas, and later send an English CV separately by email. This means that recruitment information and applicant materials are currently distributed across different channels, including announcement pages, online forms, and email attachments.

2.1.2 Perceived problems in the current workflow

Participants generally felt that the current process is not completely manual, but still fragmented. The main problem is not the absence of online tools, but the fact that different stages are handled in separate systems. For example, vacancy information is published on a portal page, application data is submitted through a form, and CVs are collected through email. Several participants said this makes the process harder to track and increases repeated manual work for both applicants and organisers.

2.2 Customer Feedback Regarding the Current Recruitment Process

A. Student / applicant-side feedback

Feedback 1

Right now the process is not exactly confusing, but it is a bit scattered. I first read the notice on the portal, then fill in the form, and then I still need to send my English CV by email. It feels like one application is split into two or three separate steps on different platforms.

Feedback 2

The vacancy notice itself is useful, but when I actually apply, I still need to switch between the portal page, the form, and my email. If one system could handle all of that together, it would definitely be more convenient.

Feedback 3

For me, the most important thing is to know whether my application has been received and what stage it is at. At the moment, after sending the form and the CV, I can only wait for a message or interview notice.

Feedback 4

If there is a skills function, I would prefer to fill it in myself in the profile. I don't think automatically extracting abilities from a CV would be very reliable, especially when different students write things in different ways.

B. Module organiser / reviewer-side feedback

Feedback 5
From the organiser side, one difficulty is that applicant information does not come in one consistent package. Basic form data and CV materials are separated, so reviewing candidates often means checking multiple sources before making a judgement.

Feedback 6
The existing form already asks students to indicate modules or areas of knowledge they have, which is useful. But if that information could be structured more clearly inside a system, it would make comparison between applicants much easier.

Feedback 7
A detailed vacancy page would be valuable, because many students apply based only on the title or a general impression. If the system clearly shows required skills, responsibilities, and expectations, that should improve the quality of applications.

Feedback 8
I would welcome a simple matching or missing-skill indicator, but only as a supporting tool. It should help us review applicants faster, not replace academic judgement.

Feedback 9
At the moment, applicants often ask repeatedly whether there will be an interview or when a result will come out. If the system could display status updates, that would reduce repetitive communication.

C. Administrative / coordination feedback

Feedback 12
The current process works, but it depends on people manually connecting information from the notice, the form, and the email submissions. A system that integrates these stages would improve both clarity and administration.

Feedback 13
Expired vacancies should not stay in the active list, but historical applications should still be retained. Administrative records are still useful after the deadline.

D. Core user needs identified

The most frequently implied needs were:

- a single place to view vacancy information,
- one integrated application process rather than “form + email”,
- clearer vacancy requirements and skill expectations,
- visibility of application status after submission,
- easier comparison of applicants for module organisers.

2.3 Interview Response Summaries

A. Summary of Responses from TA Applicants

Q1. At which step do you feel the most uncertainty or lack of information? What exactly do you wish had been clearer?

Applicants most often felt uncertain at the stage of deciding whether a vacancy was worth applying for. They said that the current announcement usually contains a lot of information, but it is not always easy to quickly identify the parts that matter most to them. What they wanted to see more clearly included the actual duties, how much time the role would take, whether the position was more technical or administrative, what prior knowledge was really expected, and whether the role was suitable for their campus and background. Several interviewees also mentioned that after submitting the application, the next stage often became unclear, especially whether their materials had been received and whether they should expect an interview.

Key insight:
Applicants do not just need "more information"; they need better-structured information for decision-making.

Q2. Do you feel that the current process requires repeated actions or switching between platforms? Which part feels most inconvenient?

Most applicants described the current process as manageable but fragmented. They noted that one application is split across several places: the vacancy is first seen on the portal, personal information is submitted through an online form, and the English CV is sent separately by email. Interviewees said the inconvenience was not that each individual step was difficult, but that they had to keep switching context and making sure nothing was missed. Some also mentioned a sense of repetition, because information already entered in the form still had to be reflected again in the CV or email process. The most inconvenient part was the lack of a single place to confirm that everything had been submitted successfully.

Key insight:
The issue is more about the lack of an integrated application flow.

Q3. If the system included a skills section in your profile, what would be the most acceptable way for you to provide that information? Why?

Applicants generally preferred profile-based skill input over automatic extraction from CVs. Many felt that the most acceptable approach would be a mixed design: a predefined list of common skill tags for quick selection, plus a small free-text field for additional skills that do not fit neatly into the preset options. Interviewees said pure free-text input can become messy and inconsistent, while fully fixed options may feel too rigid, especially for interdisciplinary or less standard backgrounds. Automatic extraction from CVs was seen as less trustworthy, mainly because students describe their abilities in different ways and were unsure whether the system would interpret them correctly.

Key insight：
A hybrid structured-skills model is more acceptable than either pure free text or CV extraction.

Q4. If the system showed a "match score" between you and a vacancy, would you trust it? Under what conditions would it feel useful rather than misleading?

Most applicants said they would not blindly trust a match score, but many were open to using it as a reference. The feature was seen as useful only if the system also explained what the score was based on, such as matched skills, missing requirements, or eligibility conditions. Interviewees were more comfortable with a simple and transparent score than with something that looked "smart" but could not be explained. Several people also said that such a score should help them decide whether to read more carefully or apply, rather than directly discouraging them from applying. A few explicitly mentioned that if the score was based only on editable profile skills and stated vacancy requirements, it would feel more fair and understandable.

Key insight：
Applicants accept matching as decision support, not as a black-box decision-maker.

Q5. What would make the application process feel more transparent or fair from a student's perspective?

Applicants repeatedly linked fairness to visibility. They wanted clearer vacancy requirements before applying, confirmation that their application had been received, and a visible status update after submission, even if the status remained "under review" for some time. Some also said that fairness was affected by whether the system made expectations explicit from the start, such as skill requirements, campus restrictions, workload, and whether prior experience would be treated as essential or only preferred. A few interviewees mentioned that seeing historical outcomes or at least a consistent review process would make the system feel more trustworthy.

Q6. If the system could only improve one thing in the first version, what should it improve first?

The most common answer was not an advanced feature, but integration. Applicants most wanted one complete workflow that combines vacancy viewing, profile submission, CV upload, and application tracking in one place. Some participants specifically prioritised application status tracking, while others focused on clearer vacancy information, but these answers still pointed back to the same core idea: the first version should make the basic application process complete, trackable, and less fragmented.

Key insight：
First-version value comes from workflow completeness, not feature novelty.

B. Summary of Responses from Module Organisers / Reviewers

Q1. Which applicant information is essential for your judgement, and which information is often missing, inconsistent, or hard to compare?

Reviewers said the most essential information usually includes the applicant's academic background, relevant module knowledge, technical or practical skills, availability, campus, prior TA or related experience, and the CV itself as supporting material. However, they also pointed out that the same information is often presented in very different ways across applicants. Some skills are implied in the CV but not stated explicitly, some experiences are described in too much detail, and some applicants provide only minimal information. This makes cross-applicant comparison difficult, even when the reviewers already know what they want to look for. Several interviewees felt that a more structured applicant profile, combined with access to the original CV, would make the review process more efficient and fair.

Key insight：
Reviewers need structured comparison data with original supporting evidence.

Q2.Which part of the current process takes the most time or creates the most repeated work for organisers?

The biggest burden mentioned by organisers was having to review applicant information across separate sources. They described repeatedly moving between form responses, emailed CV attachments, and the original vacancy requirements just to build a full picture of one candidate. This was especially inefficient when many applicants applied for the same role. Some interviewees also mentioned the communication burden after submission, because students often ask for updates individually.

Key insight：
The organiser-side inefficiency comes from data scattering and repeated reconstruction of context.

Q3. When publishing a vacancy, which information must be clearly stated to reduce unsuitable applications?

Organisers emphasised that the vacancy should clearly state the specific module, the actual type of work involved, expected responsibilities, required and preferred skills, campus constraints, workload, and whether the role is suitable only for certain student levels or backgrounds. Interviewees noted that unsuitable applications often do not come from carelessness, but from ambiguity: students may see a course title, feel generally interested, and apply without fully understanding what the role really requires.

Key insight：

A good vacancy page is not just informative; it acts as pre-screening through clarity.

Q4. If the system could help you compare applicants better, what kind of comparison would be most useful: skills, workload, experience, campus, module familiarity, or something else?

Reviewers did not point to a single factor; instead, they wanted comparison across a small set of clearly organised dimensions. The most frequently mentioned were relevant skills, familiarity with the module content, previous TA or related teaching experience, campus suitability, and current workload or time availability. Some interviewees added that they did not necessarily want the system to rank people automatically, but they did want it to make these dimensions easier to view side by side. In other words, the value lies less in fully automated selection and more in improving reviewer judgement through structured comparison.

Key insight：

What reviewers need most is comparison support, not automatic ranking.

Q5. If the first release must stay simple, which organiser-side functions are absolutely necessary?

For the first version, organisers consistently prioritised three functions: posting vacancies in a structured way, viewing applicant information in one place, and updating application outcomes or statuses centrally. They generally regarded advanced features such as matching scores or missing-skill indicators as optional enhancements rather than first-release essentials. The overall preference was for a simple but reliable organiser-side workflow that reduces manual coordination and makes decision-making easier.

Key insight：

For organisers, first-version success depends on centralised management and review efficiency.

Summary: Across both applicant-side and organiser-side interviews, a common pattern emerged: the current process already has online components, but it still feels fragmented because information, materials, and status updates are not integrated into a single workflow. Applicants mainly care about clarity, visibility, and confidence that their application is progressing properly, while organisers care about structured information and easier comparison. Both groups were open to lightweight skills or matching features, but only when those features were transparent, explainable, and clearly secondary to the core recruitment workflow.

2.4 Pilot Survey Results

Total respondents： 24

Q1. What best describes your role?

Role	Count	Percentage
Student / potential TA applicant	12	50.0%
Current / former TA	7	29.2%
College staff / module organiser	3	12.5%
Administrative / coordination role	2	8.3%

The sample was mainly composed of student-side participants, with smaller organiser-side and coordination-side input. This reflects the fact that the primary users of the application process are students, while organiser-side views were still included for balance.

Q2. Have you ever participated in an International School TA application or recruitment process?

Option	Count	Percentage
Yes	14	58.3%
No, but I am familiar with it	6	25.0%
No	4	16.7%

Q3. Which parts of the current TA recruitment process do you think are inconvenient or inefficient? (Multiple choice)

Option	Count	Percentage
Vacancy information is spread across different places	16	66.7%
Need to switch between portal, form, and email	**18**	**75.0%**

Option	Count	Percentage
Applicant information is submitted repeatedly	11	45.8%
Vacancy requirements are not clear enough	13	54.2%
Hard to know application status after submission	**17**	**70.8%**
Hard for organisers to compare applicants efficiently	9	37.5%
No major issue	1	4.2%
Other	2	8.3%

The most commonly reported issues were platform switching, unclear status visibility after submission, and fragmented vacancy information. This supports the need for a more integrated application workflow and status-tracking mechanism.

Q4. In your opinion, which information is most important before applying for a TA vacancy? (Choose up to 3)

Option	Count	Percentage
Module / course name	11	45.8%
Organiser / teacher	6	25.0%
Duties and responsibilities	**17**	**70.8%**
Required skills / knowledge	**18**	**75.0%**
Preferred experience	7	29.2%
Workload / time commitment	15	62.5%
Campus requirement	10	41.7%
Payment / reward	8	33.3%
Application deadline	11	45.8%

Respondents cared most about required skills, duties, and workload before deciding whether to apply. This suggests that a vacancy detail page should focus on practical decision-making information rather than only basic course labels.

Q5. Which functions do you think are most important in a TA recruitment system? (Multiple choice)

Function	Count	Percentage
View vacancies in one place	**18**	**75.0%**
View detailed vacancy information	**19**	**79.2%**
Submit application within the system	17	70.8%
Upload CV within the system	15	62.5%
Track application status	**20**	**83.3%**
Search vacancies by keyword	12	50.0%
Filter vacancies	10	41.7%
Structured applicant profile	11	45.8%
Applicant comparison support for organisers	8	33.3%
Workload overview	6	25.0%
Skill matching / recommendation	7	29.2%

Status tracking, detailed vacancy information, integrated vacancy browsing, and in-system application submission were the most valued functions. Advanced features such as workload overview and skill matching were selected less frequently.

Q6. If the first version must stay simple, which three functions should be prioritised first?

(Choose up to 3)

Function	Count	Percentage
View vacancies in one place	14	58.3%
View detailed vacancy information	**15**	**62.5%**
Submit application within the system	14	58.3%
Upload CV within the system	8	33.3%
Track application status	**16**	**66.7%**
Search vacancies by keyword	6	25.0%

Function	Count	Percentage
Filter vacancies	4	16.7%
Structured applicant profile	5	20.8%
Applicant comparison support for organisers	4	16.7%
Workload overview	2	8.3%
Skill matching / recommendation	2	8.3%

Respondents focused on vacancy details, integrated application flow, and status tracking, rather than advanced or analytics-oriented functions.

Q7. How useful would the following features be?(1–5 scale)

Feature	Average Score
Detailed vacancy page	**4.5**
Application status tracking	**4.6**
Vacancy search	3.9
Vacancy filtering	3.7
Structured applicant profile	3.8
Skill matching	3.2
Missing-skill hints for organisers	3.1

Detailed vacancy pages and application status tracking received the highest usefulness ratings, while skill-based intelligent features were seen as moderately useful rather than essential.

Q8. If the system shows a match score between an applicant and a vacancy, what would make it feel trustworthy? (Multiple choice)

Option	Count	Percentage
It explains which skills matched	18	75.0%
It shows which requirements are missing	16	66.7%

Option	Count	Percentage
It is based on profile information I can edit	15	62.5%
It is only used as a reference, not a final decision	19	79.2%
I would not trust such a feature	3	12.5%

Respondents were more willing to accept a match score when it was explainable, editable in terms of input data, and clearly positioned as reference rather than automated decision-making.

Summary: The survey results showed that the most strongly preferred first-version functions were application status tracking, detailed vacancy information, integrated vacancy browsing, and in-system application submission. By contrast, skill matching and related intelligent features were seen as potentially useful but clearly lower in priority.